

UX CASE STUDIES STUDENT PORTFOLIO

Project 1 To Do App

Overview

The to do app helps users track tasks and manage daily responsibilities. The app includes ways to add, complete, and organize tasks. The design focuses on simplicity and ease of use.

Problem

Task management apps often become too complex. Too many features can overwhelm users who just want to quickly add and check off tasks. Simple actions should not require multiple steps or hidden menus.

Research

Task management apps across different platforms share common issues. The main screen often shows many elements at once including filters, sort options, and project lists before users see their actual tasks. The add task button is sometimes placed in a corner that is hard to reach. Some apps require multiple taps just to enter a basic task. Other apps show every possible field at once including notes, due dates, repeat settings, priority levels, tags, and attachments. This overwhelms users who just want to write a simple reminder. Successful task apps tend to show less information by default and prioritize the add and complete actions above everything else.

Constraints

Mobile screens only. No development phase. Design stops at high fidelity mockups.

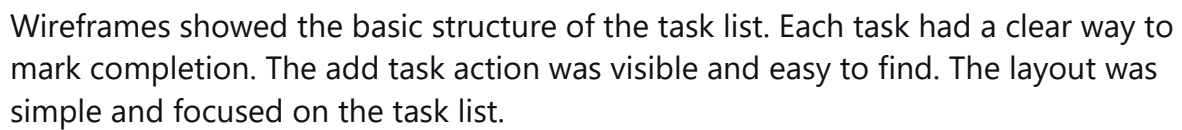
Design Decisions

- Buttons and tap targets were made large enough for easy pressing
- The add task action was placed where thumbs can reach comfortably
- Visible options were reduced to only the most essential actions
- Spacing was increased to reduce visual clutter
- Colors help communicate task priority and completion status

UX Laws Applied

- Hick's Law – reducing the number of choices makes decisions faster
- Fitts's Law – larger buttons are easier and quicker to click
- Gestalt – visually grouping related elements helps users understand relationships
- Von Restorff – making important actions look different helps them stand out

- ## Wireframes

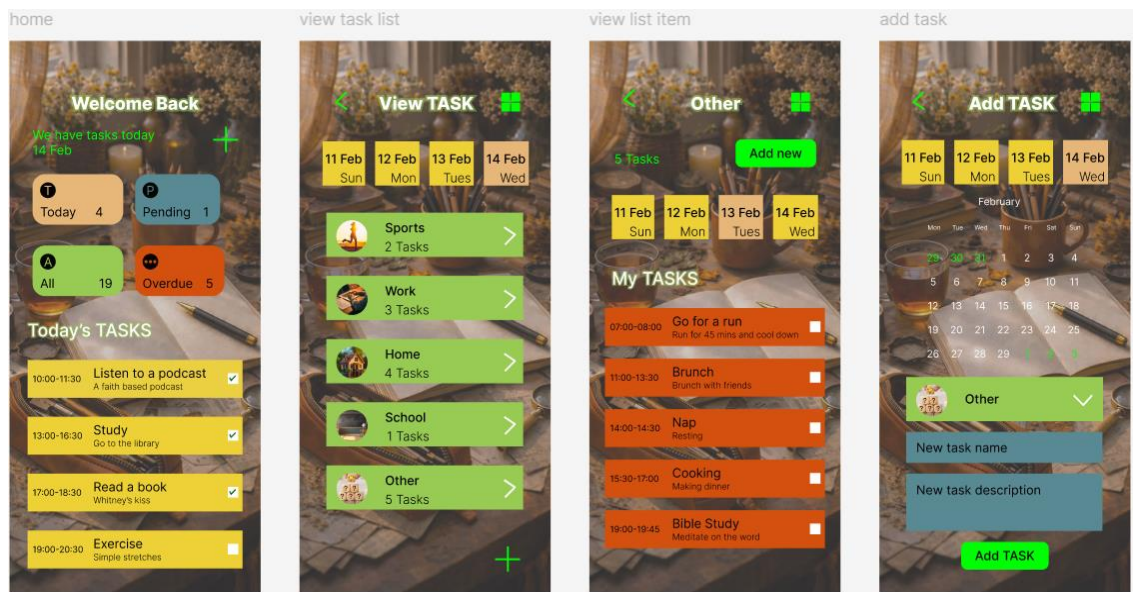


The diagram illustrates the user flow between four mobile app screens:

- home:** Displays a welcome message, task counts (Today: 4, Pending: 1), and a list of today's tasks (Listen to a podcast, Study, Read a book, Exercise).
- view task list:** Displays a calendar view and a list of tasks categorized by location (Sports, Work, Home, School, Other).
- view list item:** Displays details for a specific task (Go for a run) and a list of sub-tasks (Brunch, Nap, Cooking, Bible Study).
- add task:** Displays a calendar view and a form to add a new task (New task name, New task description).

Blue arrows indicate the flow of user interaction between these screens, showing how users can navigate from the home screen to view tasks, add new tasks, and view details of specific tasks.

Final UI



The final design keeps the task list clean and readable. Priority levels are shown through visual differences. The layout uses plenty of space so nothing feels crowded.

What Worked

Clear and easy to use with color coded tasks.

Improvements

Add reminders and usability testing.

Reflection

The biggest challenge was deciding what features to include and what to leave out. A simpler app is not a worse app. Good design often means removing things rather than adding them.

Next Steps

- Add optional reminder notifications for time sensitive tasks
- Test color choices with a small group of people
- Consider adding a dark mode option

Project 2 Music Player

Overview

The music player app lets users control playback, browse songs, and manage playlists. The design focuses on making playback controls easy to use without looking at the screen.

Problem

Music apps can feel cluttered and hard to use while walking, driving, or exercising. Small buttons make it difficult to tap accurately. Dense menus hide common actions behind multiple steps.

Research

Music playback interfaces across different platforms share similar patterns but also have frustrating differences. The play button is usually the largest element on the screen, but its size compared to skip buttons varies between apps. Some apps make all three buttons the same size, which removes visual hierarchy. Progress bars are often too small to grab accurately. The now playing bar at the bottom of many music apps can be too small to tap reliably. Some apps require users to leave the player screen just to see the upcoming playlist. The most usable music apps have simple now playing screens with large buttons and minimal text while hiding extra features behind one clear action.

Constraints

Design only. Focus on one handed use.

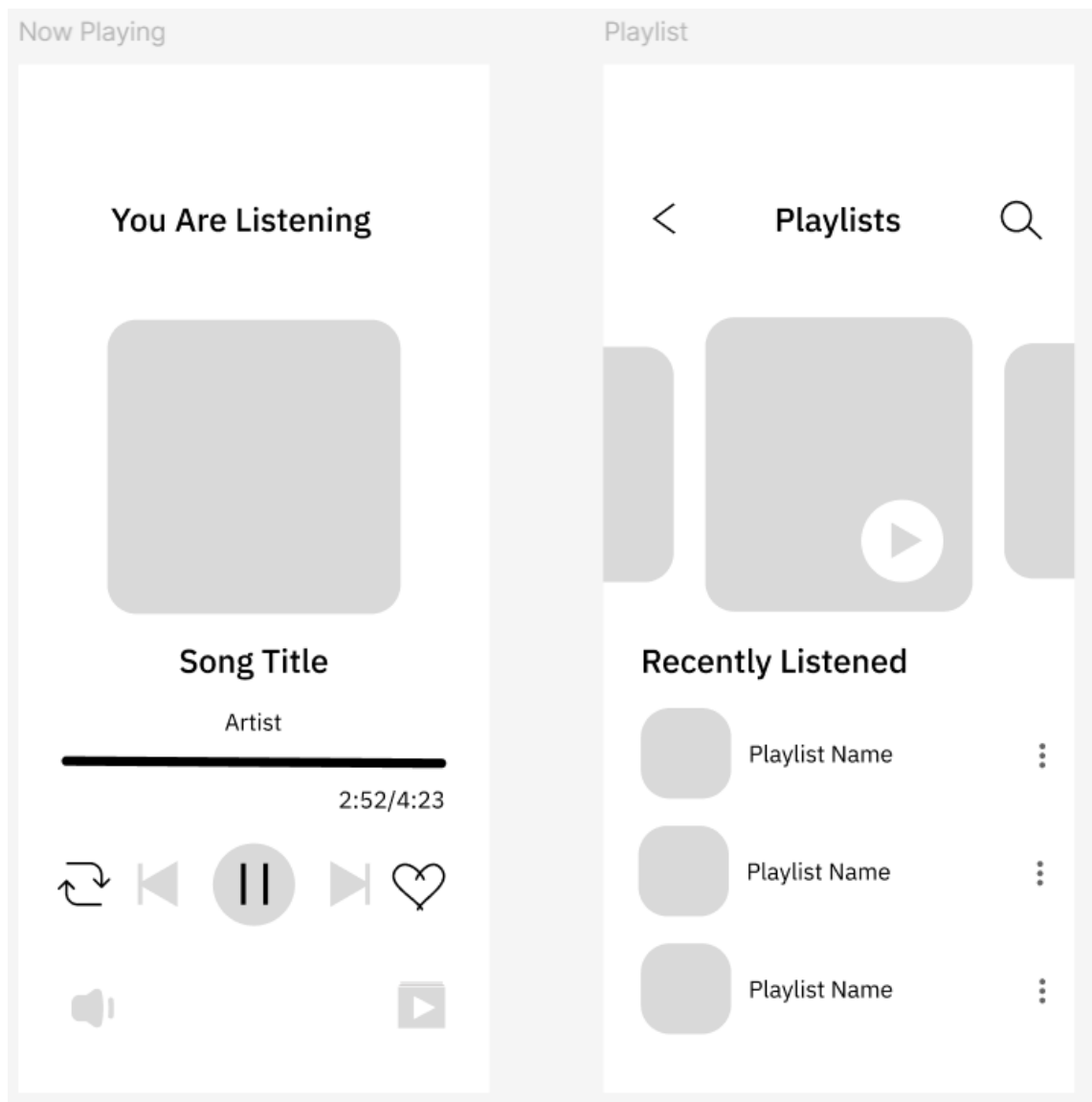
Design Decisions

- The play button was made larger than the skip buttons so it stands out
- All playback controls are placed within easy reach of the thumb

UX Laws Applied

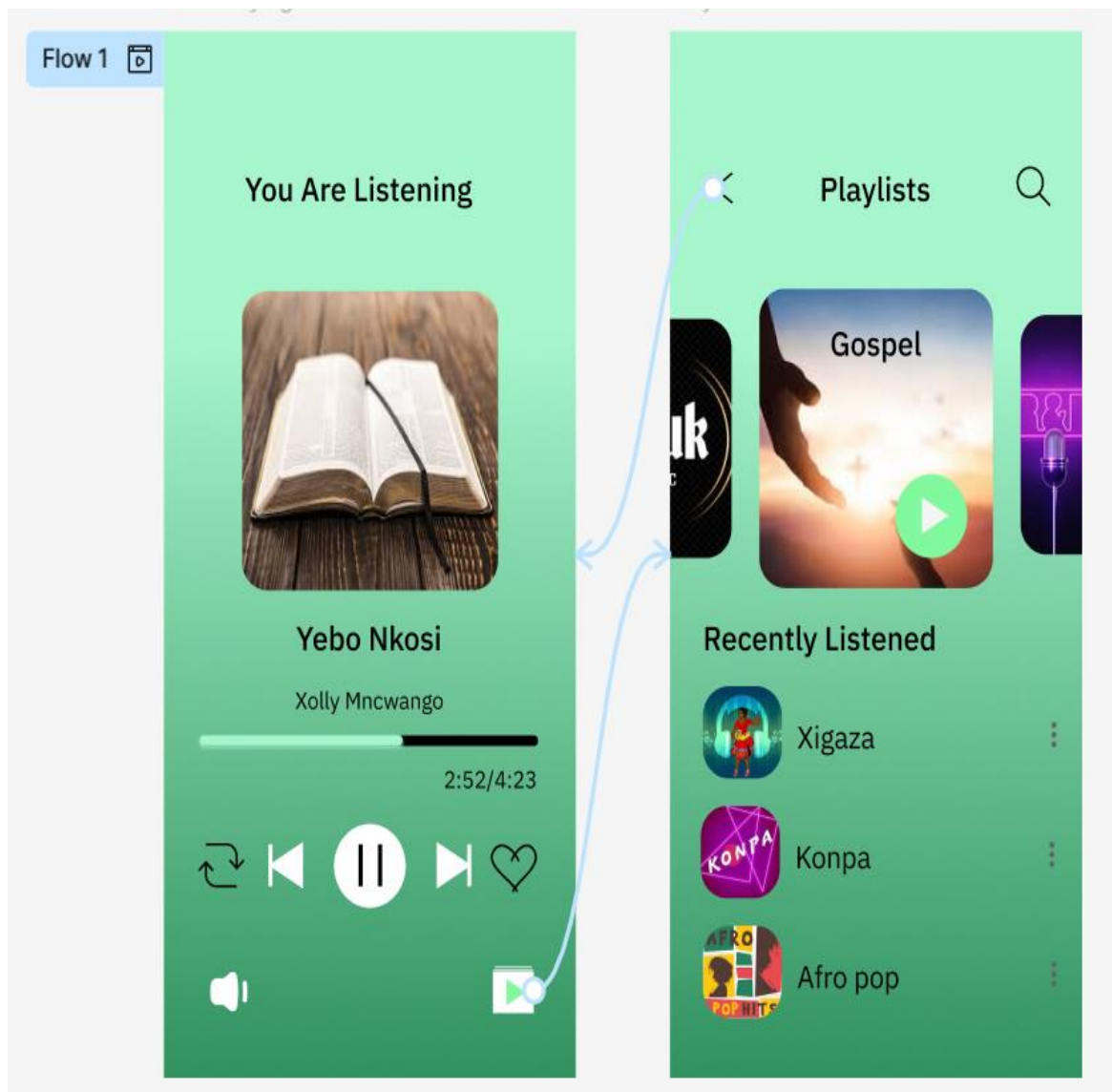
- Fitts's Law – larger buttons are easier and quicker to press
- Hick's Law – fewer visible options reduces decision time
- Gestalt – grouping related controls helps users find them faster
- Von Restorff – making the play button look different draws attention to it

Wireframes



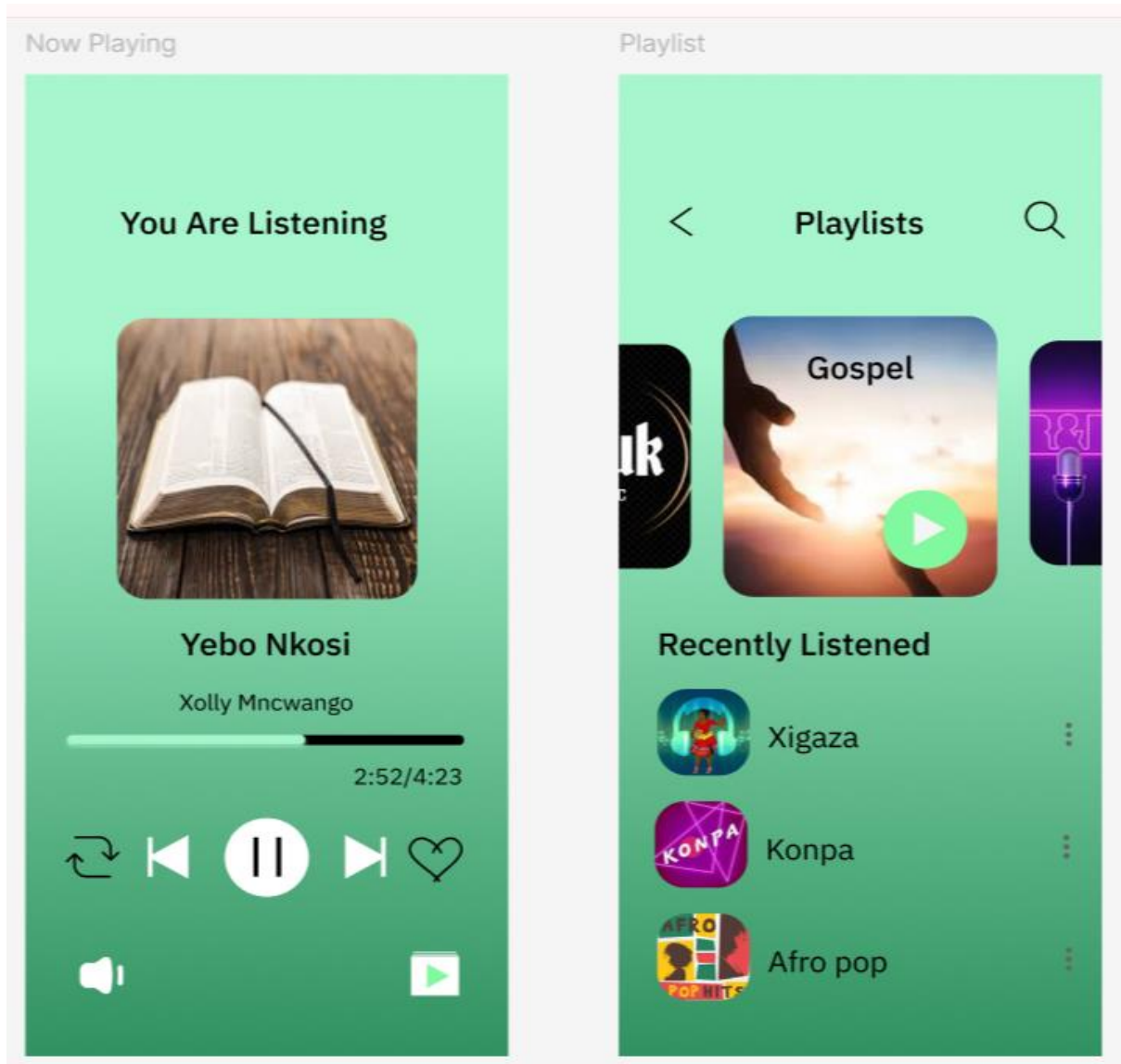
Wireframes shows playback controls are placed in a horizontal row. The playlist is located nearby but not cluttering the main view.

Prototype



The prototype added interactive states for all buttons.

Final UI



The final design keeps the main player screen clean and focused on the current song. The playlist is easy to access. A playlist at the bottom provides continuous control.

What Worked

Simple navigation and clear playback controls.

Improvements

Add search and playlist customization.

Reflection

Finding the right balance between minimalism and usefulness was the main challenge. A very simple player looks nice but may lack needed features. The best solution gives users what they need without showing everything at once.

Next Steps

- Add search functionality to find songs quickly
 - Let users customize colors based on album art
 - Test button placement with people who have different hand sizes
-

Project 3 Website Design

Overview

A website presents information to users in a clear and organized way. The design focuses on helping users find what they need quickly.

Problem

Websites often show too much content at once. Dense layouts and unclear navigation cause users to leave before finding what they came for. A well organized site respects the user's time and attention.

Research

How people read websites has been studied extensively. Users do not read line by line. They scan for keywords and visual cues. If they do not find what they need within a few seconds, they leave. The most common scanning pattern is the F pattern where users look across the top, down the left side, across the middle, and down the left side again. Another common pattern is the Z pattern across the top, diagonally down, and across the bottom. Important content needs to be placed where eyes naturally go first. Despite this knowledge, many websites continue to use dense layouts with carousels, popups, and auto playing videos that users ignore or find annoying.

Constraints

Desktop only. Visual design only. No code implementation.

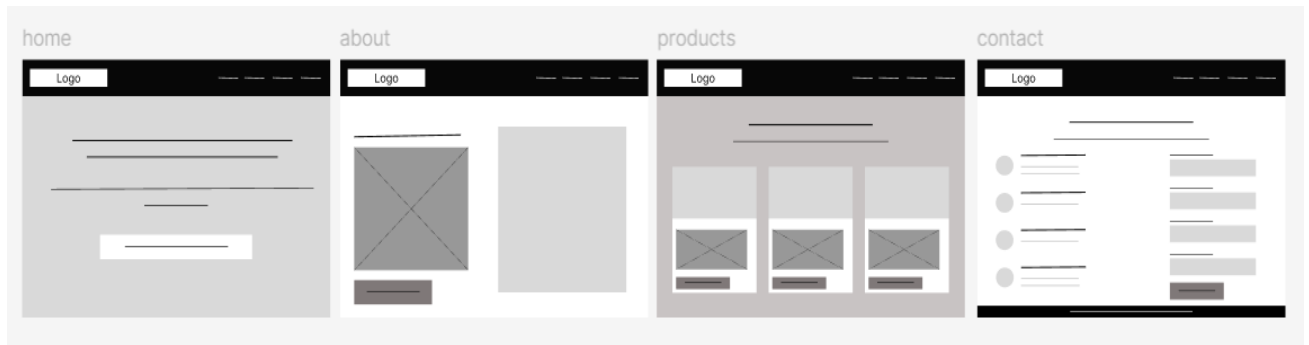
Design Decisions

- Important content is placed where users look first
- Related information is grouped together visually
- Navigation follows familiar patterns that users already understand
- Unnecessary elements like carousels and popups are not used

UX Laws Applied

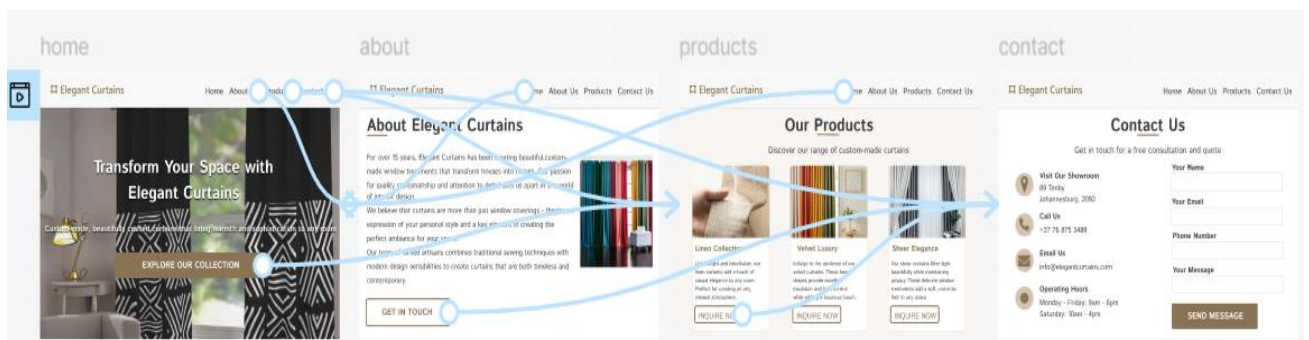
- F Pattern – users scan pages in predictable F shaped paths
- Gestalt – grouping related content helps users process information faster
- Jakob's Law – familiar layouts reduce the learning curve
- Cognitive Load – removing clutter reduces mental effort

Wireframes



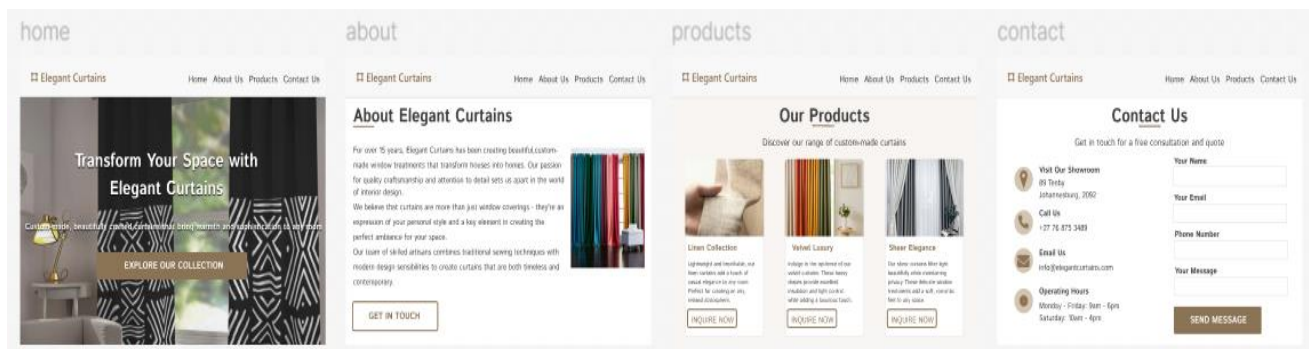
Wireframes showed the basic page structure including header, content area, and footer. Navigation was placed at the top.

Prototype



In the prototype clicking navigation items showed different sections of the page. The layout remained consistent across all pages.

Final UI



The final design uses clear visual hierarchy so users know where to look first. The navigation is easy to find and use. The overall layout feels open and readable.

What Worked

Clear hierarchy and easy navigation.

Improvements

Improve responsiveness and accessibility.

Reflection

The hardest part was deciding what to have in the pages. Every piece of content seemed important, but too much content makes nothing stand out.

Next Steps

- Improve accessibility by checking color contrast and adding focus states for keyboard navigation
- Add a way for users to jump between sections on long pages
- Test the navigation with a few people to ensure it is easy to use